Journal of Chromatography A, 1442 (2016) 33–41

Contents lists available at ScienceDirect

Journal of Chromatography A

journal homepage: www.elsevier.com/locate/chroma

Comprehensive two-dimensional liquid chromatographic analysis of poloxamers

Muhammad Imran Malik[a,b,**], Sanghoon Lee[a], Taihyun Chang[a,*]

[a] Division of Advanced Materials Science and Department of Chemistry, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea

[b] H.E.J. Research Institute of Chemistry, International Centre for Chemical and Biological Sciences (ICCBS), University of Karachi, Karachi 75270, Pakistan

ARTICLE INFO

Article history:
Received 4 December 2015
Received in revised form 13 February 2016
Accepted 4 March 2016
Available online 10 March 2016

Keywords:
Poloxamers
Liquid chromatography at critical conditions
Two-dimensional liquid chromatography
Interaction chromatography (IC)
IC–IC coupling

ABSTRACT

Poloxamers are low molar mass triblock copolymers of poly(ethylene oxide) (PEO) and poly(propylene oxide) (PPO), having number of applications as non-ionic surfactants. Comprehensive one and two-dimensional liquid chromatographic (LC) analysis of these materials is proposed in this study. The separation of oligomers of both types (PEO and PPO) is demonstrated for several commercial poloxamers. This is accomplished at the critical conditions for one of the block while interaction for the other block. Reversed phase LC at CAP of PEO allowed for oligomeric separation of triblock copolymers with regard to PPO block whereas normal phase LC at CAP of PPO renders oligomeric separation with respect to PEO block. The oligomeric separation with regard to PEO and PPO are coupled online (comprehensive 2D-LC) to reveal two-dimensional contour plots by unconventional 2D IC × IC (interaction chromatography) coupling. The study provides chemical composition mapping of both PEO and PPO, equivalent to combined molar mass and chemical composition mapping for several commercial poloxamers.

1. Introduction

Poloxamers are one of the very important industrial polymers and are available under different trade names such as Synperonics, Pluronics, and Kolliphor etc. These are basically triblock copolymers composed of poly(ethylene oxide) (PEO) and poly(propylene oxide) (PPO) blocks that are arranged in PEO-*b*-PPO-*b*-PEO or PPO-*b*-PEO-*b*-PPO sequences. The lengths and arrangement of different blocks are varied as per requirement of the applications. The PEO block serves as hydrophilic block whereas PPO block has hydrophobic nature. Therefore, poloxamers tend to form micellar nanostructures when dropped in a solvent (such as water) that has different selectivity for PEO and PPO. Obviously the stability, size and shape of micelles strongly depend upon the length of the blocks and their arrangements. The critical micelle concentration, size, and shape of these micelles in turn determine the possibilities of various applications of these fascinating materials. The poloxamers are used as one of the most important non-ionic surfactants in many industrial applications such as cosmetics, pharmaceuticals, drug delivery processes, and material science, to name a few [1,2]. The size, shape, and critical micelle concentration of these micelles are tunable through synthetic chemistry of the molecules. The overall molar mass and its distribution, molar mass distribution of individual blocks, chemical composition distribution, and block arrangement strongly influence the physical properties, and thus the practical performance of poloxamers. The presence and extent of side product homo-polymers also influence the performance properties of the products significantly [3–6]. Meticulous characterization of polymeric materials with regard to above mentioned aspects has become a vital part of development. Despite commercial availability of several grades of poloxamers and unending list of their applications, no comprehensive characterization method has been proposed yet.

Synthetic block copolymers are heterogeneous in nature with distributions in molar mass, tacticity, architectures, topology and compositions etc. The complete analysis of block copolymers is not a trivial task as these distributions always lie on top of each other. The presence of homo-polymer side products adds more to the complexity of the samples. Different modes of liquid chromatography of polymers can reveal different information [7–12]. Size exclusion chromatography (SEC) being most widely used mode of LC of polymers separates with regard to the size of the polymer

* Corresponding author at: Division of Advanced Materials Science and Department of Chemistry, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea.

** Co-corresponding author.

E-mail addresses: mimran.malik@iccs.edu (M.I. Malik), tc@postech.ac.kr (T. Chang).

http://dx.doi.org/10.1016/j.chroma.2016.03.008
0021-9673/

species in dilute solution [13]. Solvent gradient and temperature gradient LC have been used for separation of polymers with regard to composition and molar mass. The gradient modes of LC of polymers generally start in the mobile phase composition and/or temperature at which polymers have strong adsorptive interaction with the stationary phase. During the run the interaction strength is decreased slowly by continuous change in the mobile phase composition and/or temperature. Liquid chromatography at critical conditions (LCCC) allows separation of the polymers with regard to chemical composition by making one of the block chromatographically invisible, that results in predominant influence of noncritical block on chromatographic retention. This special combination of mobile phase-stationary phase-temperature-analyte is often termed as critical adsorption point (CAP) for that particular polymer. The chromatographic invisibility for a specific polymer is achieved under carefully established balance between interaction (enthalpy driven) and exclusion (entropy controlled) so that both effects cancel each other, and polymer species elute near the void volume of the column independent of the molar mass. The critical conditions are strongly affected by changes in mobile phase composition and temperature. Therefore, LCCC is intrinsically an isocratic mode of LC of polymers. LCCC-IC has been used for analysis of functional polymers (with regard to functionality) or block copolymers if the interacting segment is narrowly dispersed and of equitably low interaction with the stationary phase [5,14–18].

For block copolymers, the noncritical block may be excluded and elute ahead of the homo-polymers under CAP or have strong interaction with the stationary phase and thus elute later [19,20]. For high molar mass samples, the most viable approach is to elute the noncritical block in exclusion regime [10,15,21–24]. Although the LCCC-SEC approach provides reasonable estimates of molar mass distribution of noncritical block, the accuracy of the technique decreases as the relative length of the block under CAP gets larger [25–27]. If the noncritical block has higher molar mass and is in interaction regime, it is generally not possible to analyze it at critical conditions because the polymers might adsorb irreversibly on the stationary phase. Gradient modes of LC of polymers are the methods of choice in these situations except for a few special exceptions if the interactive block is very much alike and/or having very low interaction with the stationary phase at CAP of the other segment [27–30].

In case of low molar mass samples such as poloxamers, exclusion from the stationary phase could not provide detailed information except the size of the polymer species in the solution. Therefore, a very fascinating situation could be critical conditions of one block and interaction of the other block that leads to oligomeric separation of noncritical block [5,17,31–33]. Jandera and coworkers reported on selectivity of several different mobile phases, stationary phases, temperature and gradients on resolution of block copolymers of PEO and PPO [34–36]. This combination is not easy to achieve in general. A gradient of two critical solvents (critical chromatography in ternary solvent [37]) that has different strength for the noncritical block can be another useful approach [38]. In principle, if the noncritical block is in exclusion regime, the retention volume is only weakly influenced by the architecture of polymer [15,19] (such as PEO-*b*-PPO diblock, PEO-*b*-PPO-*b*-PEO or PPO-*b*-PEO-*b*-PPO triblock copolymers). On the other hand, if noncritical block is in interaction regime, diblock copolymers can be well separated at critical conditions of either PEO or PPO [31]. However, in case of triblock copolymers, if the outer block is under CAP it shall show similar chromatographic behavior as that of a diblock copolymer but the retention behavior could be somewhat different in case of inner critical block [36,39,40]. We shall demonstrate in this study that, by carefully selecting conditions, an excellent oligomeric separation with both inner and outer critical blocks could be achieved.

Finally, coupling of two LCCC methods where the noncritical block is in interaction regime can reveal a two-dimensional mapping with regard to both molar mass and chemical composition of the block copolymers. SEC is the most widely used mode of LC for second dimension in two-dimensional LC (2D-LC) of polymers [8,20,23,41–45]. In this case all the polymers should elute before the void volume of the column and analysis can be completed swiftly. Another swift and useful method for the second dimension is LCCC, particularly if both first and second dimension have the same mobile-stationary phase system without compatibility problems e.g. TGIC-LCCC [27,45]. However, if the second dimension separates in the interaction regime where polymers elute later than the void volume of column in the form of well resolved peaks, it might require fairly long time for complete analysis. The second dimension analysis in interaction regime could not be completed in the time required to fill the other loop from the first dimension effluent (a typical approach used for SEC analysis in the second dimension). In this case, a multiple trapping system where sample in the effluent from the first dimension is trapped several times into a short column of identical nature as the second dimension analysis column, can be used [33,46]. In this study, sample is trapped six times prior to its analysis in the second dimension that allowed a time of twelve minutes for analysis in the second dimension. Other problems associated with this type of IC × IC coupling could be solvent incompatibility and strength of the solvent from the first dimension for the second dimension separation system [47]. Different approaches could be used to overcome these problems that include addition of poor solvent in the first dimension effluent prior to injection to the second dimension LC [48,49] and use of trapping system. Brief description of trapping system setup is given in experimental section while further details can be found in Ref. [33,46]. There are only very few reports of analysis of block copolymers of low molar mass by IC × IC coupling [33,50–53]. None of above mentioned research articles addressed triblock copolymers that behaves differently than the diblock copolymers at the CAP, as explained earlier. There have been many reports on the separation analysis of EO/PO co-polymers, but to our knowledge only few of them address the analysis using the on-line coupling of two ICs that result in full resolution of all oligomers. Furthermore, the relative block lengths of EO-PO diblock copolymers analyzed by 2D-LC in the articles mentioned above, were rather low. In this study, we tried to extend the molar mass range of individual blocks to 1000–1500 for each block.

In this study, we propose a comprehensive analysis method for commercial poloxamers. Reversed phase (RP) LC separates with regard to PO units of the copolymers at the critical conditions of PEO. On the contrary, Normal phase (NP) LC separates with respect to EO units of the copolymers with minimal effect of PPO block at its critical conditions. The two LCCC-interaction methods are coupled online (comprehensive 2D-LC) to acquire the two-dimensional contour plots of poloxamers that maps the bivariate distribution of the copolymer with regard to both molar mass and chemical composition. The study provides a finger print of several different commercial poloxamers. MALDI-TOF MS analysis of fractions is employed to confirm the success of separation and to determine the degree of polymerization of EO and/or PO.

2. Materials and methods

2.1. Materials

PEG, PPG standards and commercial Poloxamers namely Pluronic-10R5, Pluronic-L35, Pluronic-1740, and Pluronic-17R4 were purchased from Aldrich and BASF, respectively. HPLC grade

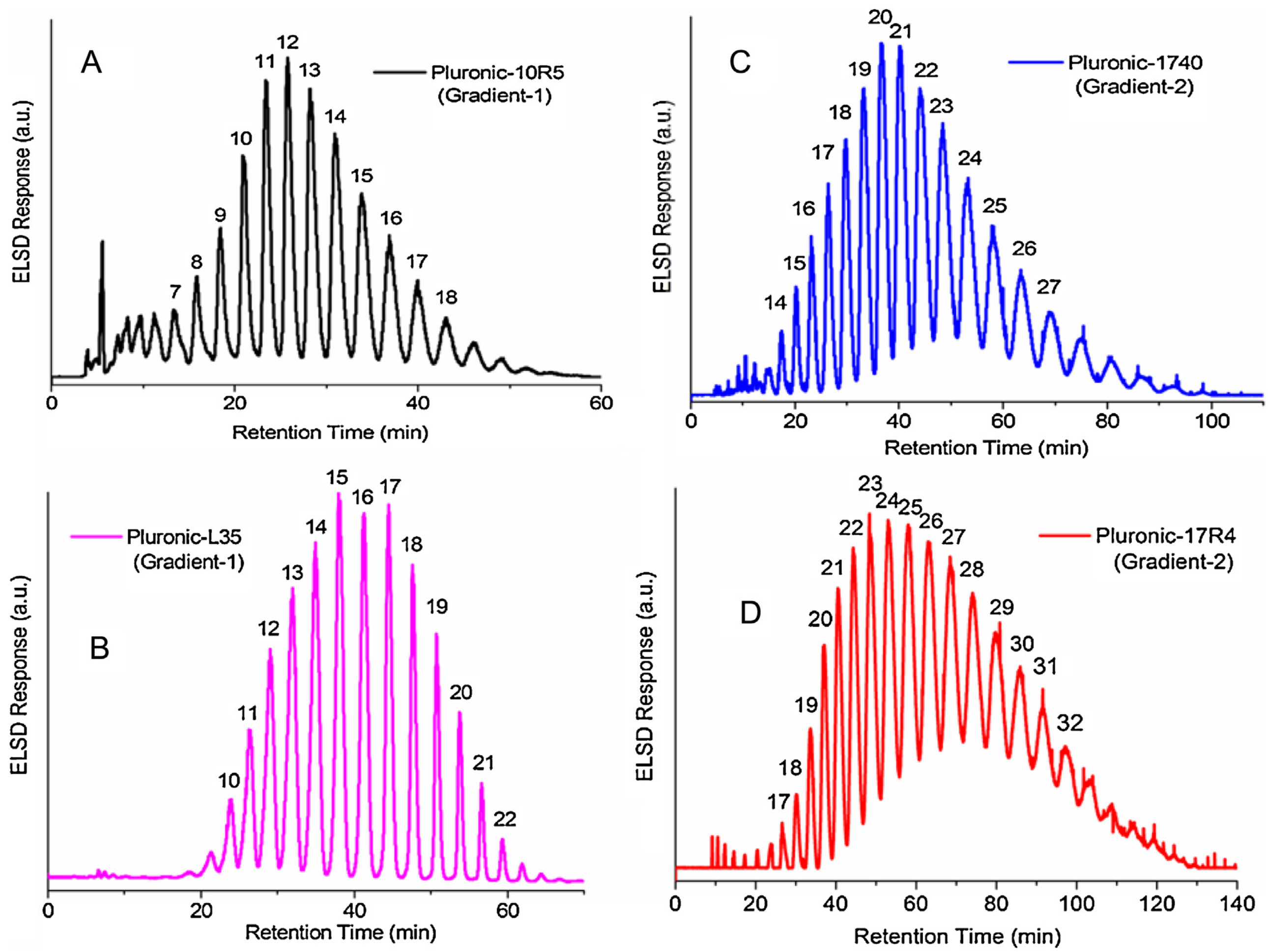

Fig. 1. Oligomeric separation of poloxamers with regard to PPO block along the critical adsorption line of PEO on Nucleosil C18, 5 μm, 300 Å 250 × 4.6 mm column at 50 °C, flow rate = 0.5 mL/min (A) Pluronic-10R5 (B) Pluronic-L35 (C) Pluronic-1740 (D) Pluronic-17R4; Solvent A: Acetone/Water = 37/63 v/v, Solvent B: Methanol/Water = 87/13 v/v; Details of gradients are given in Tables 1 and 2. Numbers given at the top of each peak represent the number of PO units as assigned by MALDI-TOF MS analysis of fractions.

acetone, methanol and acetonitrile were purchased from Samchun, Korea.

2.2. Liquid chromatography

One-dimensional liquid chromatographic analysis was performed on a modular system consisting of Shimadzu LC20AD binary pump, Rheodyne 6-port injection valve equipped with a 100 μL sample loop, and a column oven. The chromatograms were recorded by polymer laboratories PL ELS 2100, an evaporating light scattering detector. Evaporator and nebulizer temperature of detector were set at 55 and 40 °C, respectively. Filtered compressed air was used as carrier gas at a flow rate of 1.0 standard liter per minute for all measurements. Sample concentrations were 3.0 – 10.0 mg/mL.

For RPLC separations, above described HPLC system was used with a Nucleosil C18, 5 μm, 300 Å, 250 × 4.6 mm column. Various gradients of two critical solvents for PEO with different slopes were used. Solvent A was acetone/water = 37/63 v/v, while solvent B was methanol/water = 87/13 v/v. The flow rate was 0.5 mL/min and column oven was maintained at 50 °C. Specific details of gradients are given in Tables 1 and 2.

For NPLC separations, the above described HPLC system was used with a Nucleosil, 5 μm, 300 Å, 250 × 4.6 mm column. Isocratic separations at CAP of PPO were performed with mobile phase composition of acetonitrile/water = 80/20 v/v at a flow rate of 1.0 mL/min at 30 °C.

In 2D-LC, NPLC separation is coupled to RPLC separation. The detailed description of the experimental setup is given elsewhere [33,46]. Briefly, 2D-LC system consists of three HPLC pumps, a Rheodyne 6-port injection valve, two 10-port 2-position switching

Table 1
Gradient 1 (Solvent A: Acetone/Water = 37/63 v/v; Solvent B: Methanol/Water = 87/13 v/v).

Time (min)	Solvent B (%)
0	50
15	75
60	100
70	100
70.01	50

Table 2
Gradient 2 (Solvent A: Acetone/Water = 37/63 v/v; Solvent B: Methanol/Water = 87/13 v/v).

Time (min)	Solvent B (%)
0	70
30	85
120	100
150	100
150.01	70

Table 3
Information of different commercial Poloxamers as provided by producer.

Product	Pluronic-10R5	Pluronic-L35	Pluronic-1740	Pluronic-17R4
Formula	PO_8-EO_{22}-PO_8	EO_{11}-PO_{16}-EO_{11}	PO_{11}-EO_{27}-PO_{11}	PO_{14}-EO_{24}-PO_{14}
Chemical Composition (EO-PO) wt%	50–50	50–50	50–50	40–60

valves (Alltech, SelectPro), two column ovens and an ELS detector. First dimensional setup consists of a Shimadzu LC20AD binary pump, a Rheodyne 6-port injection valve equipped with a 100 µL sample loop, and a column oven. The first dimension separation was realized on a Nucleosil, 5 µm, 300 Å, 250 × 4.6 mm column at critical conditions of PPO while interaction conditions of PEO in acetonitrile/water = 84/16 v/v at a flow rate of 0.05 mL/min. The eluent from the first dimension LC is directed to a 10-port switching valve that is equipped with two 100 µL loops that are being loaded with the effluent from the first dimension alternatively. The valve switches every two minutes that is the time required to fill the loop at the flow rate of the first dimension separation (0.05 mL/min). When one of the loop is being loaded with the effluent from the first dimension, the other loop is being flushed with another mobile phase (methanol/water = 50/50 v/v) at a flow rate of 1.0 mL/min by the second pump. The effluent from the first switching valve is now directed to the second switching valve that is equipped with two short columns (Nucleosil C18, 5 µm, 500 Å, 30 × 4.6 mm) to trap the analyte under a strong adsorbing condition. The flow rate of intermediate step (1.0 mL/min) was enough to flush all the first dimension solvent from the short column. Second 10-port valve switches every 12 min that is the time required for the second dimension analysis. The analyte from the first switching valve is trapped in the short RP column for six times prior to its analysis in the second dimension at fairly critical conditions of PEO and adsorbing conditions of PPO. The analysis in second dimension was performed at a flow rate of 1.0 mL/min in methanol-water mobile phase of varying composition depending upon length of PPO block, on a Nucleosil C18, 5 µm, 500 Å, 100 × 4.6 mm column. The chromatograms were recorded by PL ELS 2100. Sample concentrations for 2D-LC analysis were kept at 20–60 mg/mL. The experimental conditions for the second dimension analysis are varied according to requirement of the PPO block lengths and are mentioned in the figure captions and discussion. The temperature of column ovens was maintained at 30 and 40–45 °C for the first and the second dimensions, respectively.

2.3. Mass spectrometry

Bruker REFLEX III was used for MALDI-TOF mass spectrometry. The instrument is equipped with a nitrogen laser (337 nm wavelength), and a time lag focusing unit. The irradiations slightly above the threshold laser power were used to generate ions. Accelerating voltage of 20 kV was applied to record positive ion spectra in reflectron mode. The system was externally calibrated with a suitable mixture of poly(ethylene glycol)s (PEG). To improve the signal-to-noise ratio, the spectra of 100–150 shots were averaged.

A solution of dithranol (30 mg/mL in THF), a solution of the analyte (fractions redissolved in THF), and a solution of CF_3COONa (10 mg/mL in THF), were mixed to prepare sample solutions in a ratio of 10:5:1.5(v/v/v). 0.5 µL of the resulting mixtures were deposited on the stainless steel sample plate and allowed to dry in air.

3. Results and discussion

The poloxamers used in this study are commercial products namely, Pluronic-10R5, Pluronic-L35, Pluronic-1740 and Pluronic-17R4. The products vary in their total molar mass, chemical

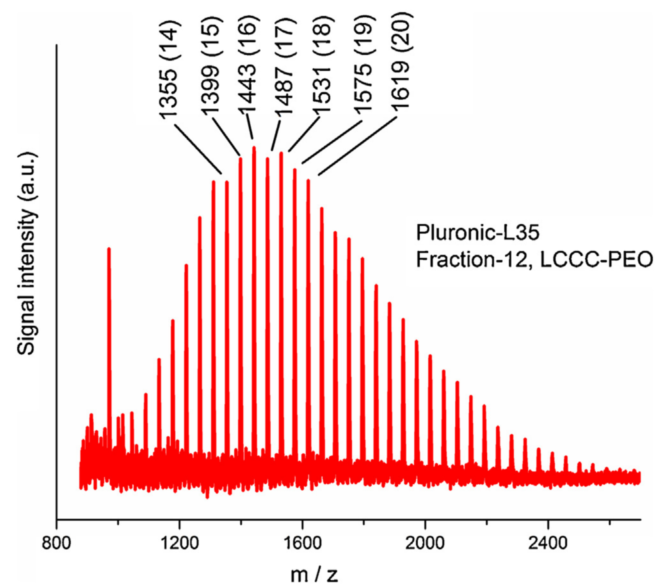

Fig. 2. MALDI-TOF MS of fraction-12 (DP_{PO} = 12) from Fig. 1B (Pluronic-L35). The *m*/*z* values are given at the top of each peak including degree of polymerization of EO in bracket; Theoretical molecular weight of co-oligomer $[(EO)_{19}\text{-}(PO)_{12} + Na]^+$ (C_{74} H_{150} O_{32} Na) = 1574.98.

composition, block sequence and block lengths as can be seen in Table 3. The first requirement for any two-dimensional LC method is the selection of appropriate conditions for both dimensions that can separate with respect to a specific property having minimal effect of other distributions. In this study, RPLC is used for separation with respect to PPO oligomers at CAP of PEO. Conversely, NPLC is employed to separate with regard to PEO block at CAP of PPO.

3.1. RPLC

In principle, critical conditions of PEO can be found on RP column and PPO should elute in interaction regime under these conditions. There are several reports on critical conditions of PEO on RP column possessing different interaction strengths for PPO with the stationary phase. Typical examples are aqueous phases of acetone, acetonitrile and methanol [31]. In acetone-water system, critical conditions of PEO can be recognized in 30–40 vol% acetone. Under these conditions PPOs interact strongly with the stationary phase and only very low oligomers could be resolved [22] (see S1 in Supplementary material). Additionally, CAP of PEO can also be realized on RP column in methanol-water mobile phase. In this mobile phase, critical conditions exist in fairly wide range of mobile phase composition (75–87 vol% methanol) [40,54]. Under these conditions, PPOs have somewhat moderate interaction with the stationary phase and oligomeric separation of PPG-1000 can be easily achieved. The molar mass range can also be extended to higher oligomers of PPG-2000, but the lower interaction with the stationary phase results in a poor resolution of lower oligomers (see S2 in Supplementary material). A very promising approach to achieve oligomeric separation from very low oligomers to fairly high oligomers of PPOs could be a gradient elution using above described critical compositions. It has been shown in previous studies that PEOs remain fairly critical along this gradient [38].

The gradient can be maneuvered according to the length and distribution of the PPO block to improve the resolution (see S3 in Supplementary material). Similar conditions have been applied for analysis of PEO-*b*-PPO diblock copolymers to achieve oligomeric separation with regard to PPO block [38]. In case of triblock copolymers, the position of critical block either in the middle or at edges may have a significant role in the chromatographic elution. If critical PEO is in the middle, the elution behavior is more affected by the length of the critical block [36,39,40].

We have now applied gradient-LCCC approach to commercial poloxamers with varying lengths and sequence of both blocks. It can be seen that carefully employed gradient can minimize effects of block sequence and baseline separation of PPO block has been achieved irrespective of its position in the block copolymer. Fig. 1 demonstrates oligomeric separation of four different poloxamers with varying lengths and sequence of the blocks. Two gradients of different slopes have been used according to the length and distribution of PPO block in the triblock copolymers (see Table 1 and 2 for details). Gradient 1 is more favorable for lower oligomers whereas gradient 2 allows elution of higher oligomers (compare product composition and sequence from Table 3). As a next step, some of the fractions were collected by heart cutting and analyzed by MALDI-TOF MS for confirmation of successful separation and assignment of the number of PO units. The fractions should have a distribution with regard to PEO block attached to a single length of PPO block. As a typical example, MS spectrum of fraction 12 of Pluronic-L35 (Fig. 1B) is shown in Fig. 2. A single series of ions with increment of 44 (formula weight of one EO unit) is obtained. The *m*/*z* values are given at the top of each peak followed by degree of polymerization of EO in bracket. The fraction has 12 PO units attached to several EO block lengths as can be seen in Fig. 2. The MALDI-TOF MS analyses of fractions confirm that triblock copolymers were separated with regard to number of PO units in the block copolymer. The numbers of PO units (given at the top of peaks in Fig. 1) are assigned with the help of MALDI-TOF MS analysis of the fractions. The product does not seem to have significant amount of PEO homo-polymers that should have been eluted near the void volume of the column, if present.

3.2. NPLC

On the contrary, NPLC can be used to achieve CAP of PPO and interaction of PEO [31]. In this study, we realized critical conditions of PPO on Nucleosil columns in acetonitrile-water mobile phase. PPOs show molar mass independent elution in the range of interest for this study (up to 2500 g/mol) in acetonitrile/water (80/20 v/v). As expected, PEOs elute in interaction regime allowing a nice oligomeric separation of PEO-1000 (see S4 in Supplementary material). These conditions have been applied for analyses of the products under study to realize separation with regard to PEO block. As mentioned earlier, the length of middle block even if it is critical can influence the elution of the triblock copolymers [36,39,40]. Fig. 3 demonstrates the separation of the same commercial poloxamers at critical conditions of PPO. The chromatograms shown in

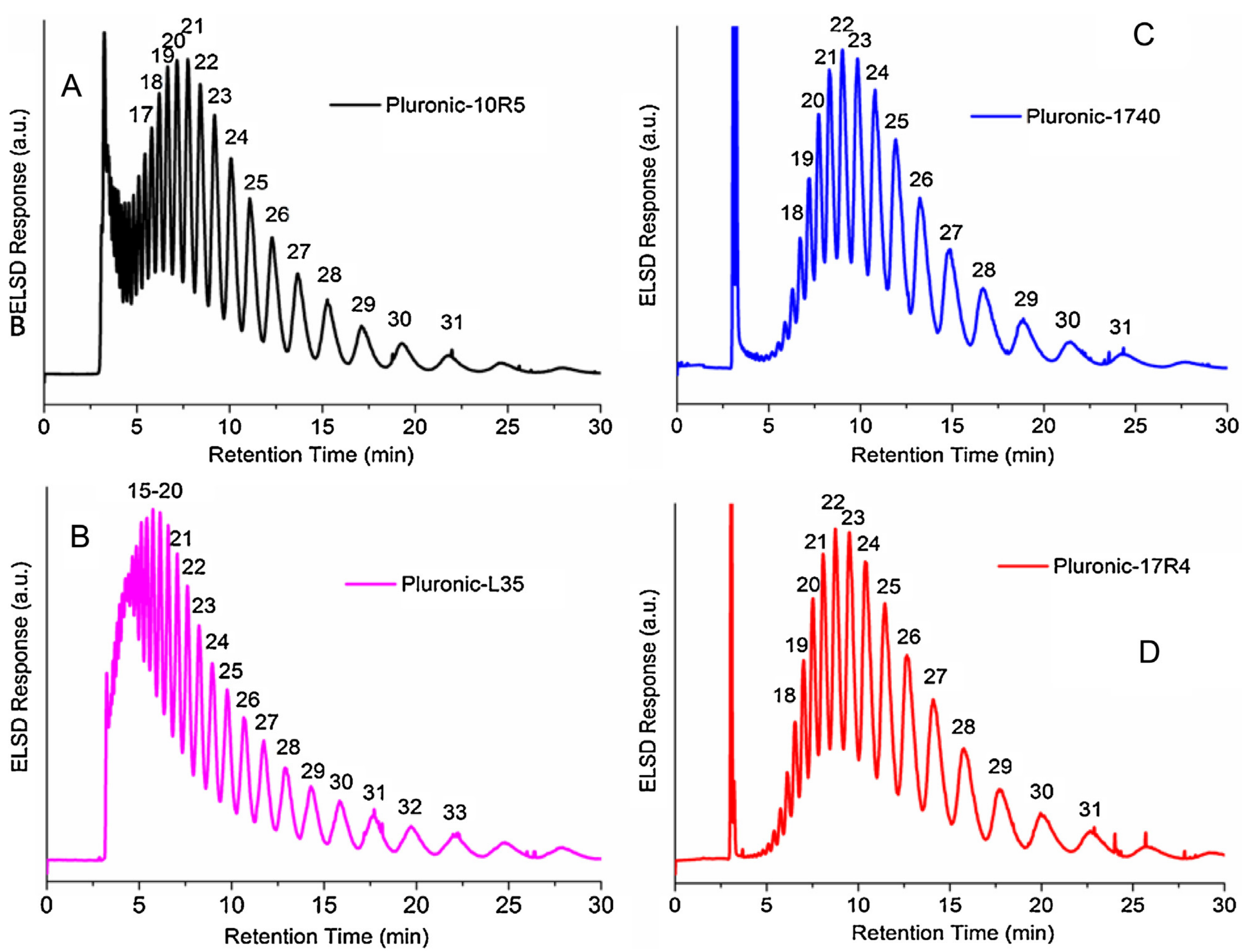

Fig. 3. Oligomeric separation of poloxamers with regard to PEO block at CAP of PPG on Nucleosil, 5 μm, 300 Å, 250 × 4.6 mm column in acetonitrile/water = 80/20 v/v at 30 °C, flow rate = 1.0 mL/min; A) Pluronic-10R5, B) Pluronic-L35, C) Pluronic-1740, D) Pluronic-17R4.

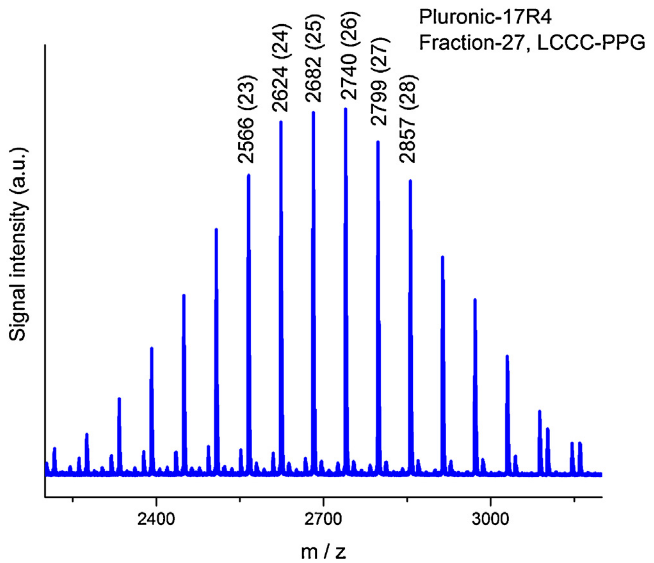

Fig. 4. MALDI-TOF MS of fraction-27 from Fig. 3D (Pluronic-17R4). The *m/z* values are given at the top of each peak including degree of polymerization of PO in bracket; Theoretical molecular weight of co-oligomer $[(EO)_{27}\text{-}(PO)_{25} + Na]^+$ (C_{129} H_{260} O_{53} Na) = 2682.45.

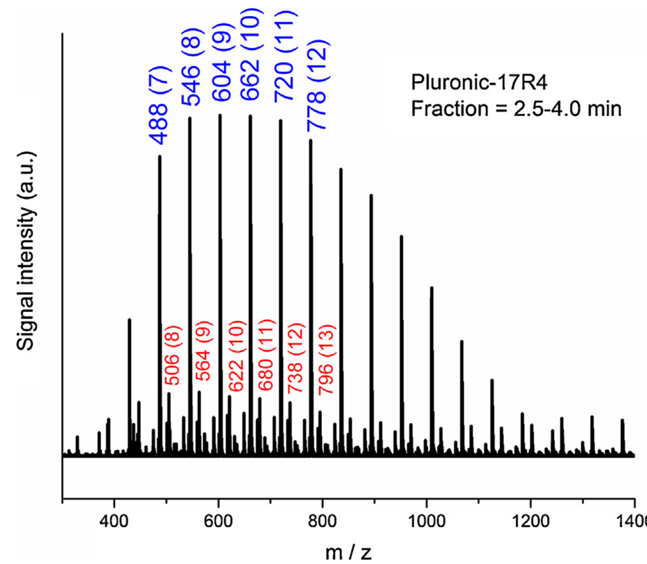

Fig. 5. MALDI-TOF MS of fraction between 2.5–4.0 min from Fig. 3D (Pluronic-17R4). The *m/z* values are given at the top of each peak including degree of polymerization of PO in bracket; Blue label (for major peaks): monoallyl PPOs, Red label (for minor peaks): PPGs; Theoretical molecular weight of monoallyl-$P(PO)_{10}$ (C_{33} H_{66} O_{11} Na) = 661.87 and PPG $(PO)_{11}$ (C_{33} H_{68} O_{12} Na) = 679.88. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 3 depict the separation with respect to the number of EO units in the triblock copolymer. The effect of the position of critical block is more pronounced in this case compared to separation of the same products with respect to the PPO block (Fig. 1). The pluronic-L35 is a product with PPO block in the center. As can be seen in Fig. 3, somewhat better looking chromatograms were obtained for the products with outer critical block compared to the one with inner critical block, nonetheless, there is not a pronounced difference in the separation of different structures (ABA vs BAB). The difference in the retention times of the oligomers of two types of structures (ABA vs BAB) is consistent with the theoretical prediction [55].

To confirm the successful separation and to assign the number of EO units to peaks, MALDI-TOF MS analyses of several fractions were conducted. Each peak in this case should be separated with regard to the number of EO units at CAP of PPO. Therefore, each peak should have a distribution in the PO block attached to a uniform PEO block. The expected MALDI-TOF MS peak pattern is a single series of ions with an increment of 58 (formula weight of a PO unit). This is really the case as can be seen in Fig. 4, a representative MALDI-TOF MS of a fraction-27 of pluronic-17R4 (from Fig. 3D). The *m/z* values of the co-oligomers are given at the top of each peak of MALDI-TOF MS spectra followed by degree of polymerization of PO in bracket. Each peak in the chromatograms shown in Fig. 3 is assigned with a number that corresponds to the number of EO units with the help of MALDI-TOF MS analysis of fractions. On the contrary to the CAP of PEO, all the products have a strong peak near the void volume of the column at the CAP of PPO. These could be PPO with allylic end group that can be formed by chain transfer to the monomer [21,54]. These PPOs with allylic end group can be initiated at any time during polymerization and propagate only at one side of the chain, therefore, expected to have lower molecular weight in comparison to the PPO block in the block copolymer, as can be seen in the spectrum. The MALDI-TOF MS analysis of the peak eluting between 2.5–4.0 min for pluronic-17R4 confirms the above assumptions (see Fig. 5). Additionally, traces of PPGs are also detected as a minor series in MALDI-TOF spectrum of this fraction that can be formed due to presence of any moisture content in the reaction mixture.

3.3. Two-dimensional LC analysis

The success of two-dimensional IC × IC LC system depends upon several factors namely, miscibility of the solvents, strength of first dimensional solvent for second dimensional analysis, trapping efficiency in case of trapping interface, and speed without loss of much resolution etc. The conditions applied for 2D analysis could be different from 1D analysis in order to achieve optimal 2D separation at flow rates, temperature and column dimensions. First dimensional analysis has to be very slow where as second dimension analysis has to be fast enough to be completed in minimum possible time. In case of IC × IC coupling, the analysis in the second dimension usually takes longer than time required to fill the other sample loop at a flow rate of first dimension, therefore, it is often inevitable to trap the first dimension analyte for several times in order to analyze the sample in a comprehensive manner. Additionally, trapping effectively concentrates the sample prior to second dimension analysis. Trapping efficiency is also a very crucial factor. Analytes should have been trapped during multiple trappings at the same point of the short column and should not be moved by next few injections of first dimensional solvent. After evaluation of these basic tests, we decided to couple NP separation to RP. The first dimension for all the products has been performed at fairly critical conditions of PPO in acetonitrile/water (84/16 v/v) on a bare silica column and is given on y-axis. The second dimensional analysis is performed on a C18 silica column with pore diameter of 500 Å, represented by x-axis. Two short columns of identical stationary phase (silica C18, 5 μm, 500 Å, 30 × 4.6 mm) are used for trapping. The mobile phase composition, flow rate, and column oven temperature for the second dimension analysis were varied as per requirements of the length of PPO block of the triblock copolymer. The separation with regard to number of EO units is shown by y-axis and PO block length is given by x-axis.

Fig. 6C represents the 2D IC × IC mapping of pluronic-1740. In this case, methanol/water (84/16 v/v) is used at a flow rate of 1.1 mL/min for second dimension analysis. For pluronic-10R5 (Fig. 6A), pluronic-L35 (Fig. 6B), and pluronic-17R4 (Fig. 6D) methanol-water mobile phase of composotions 80/20 v/v, 75/25

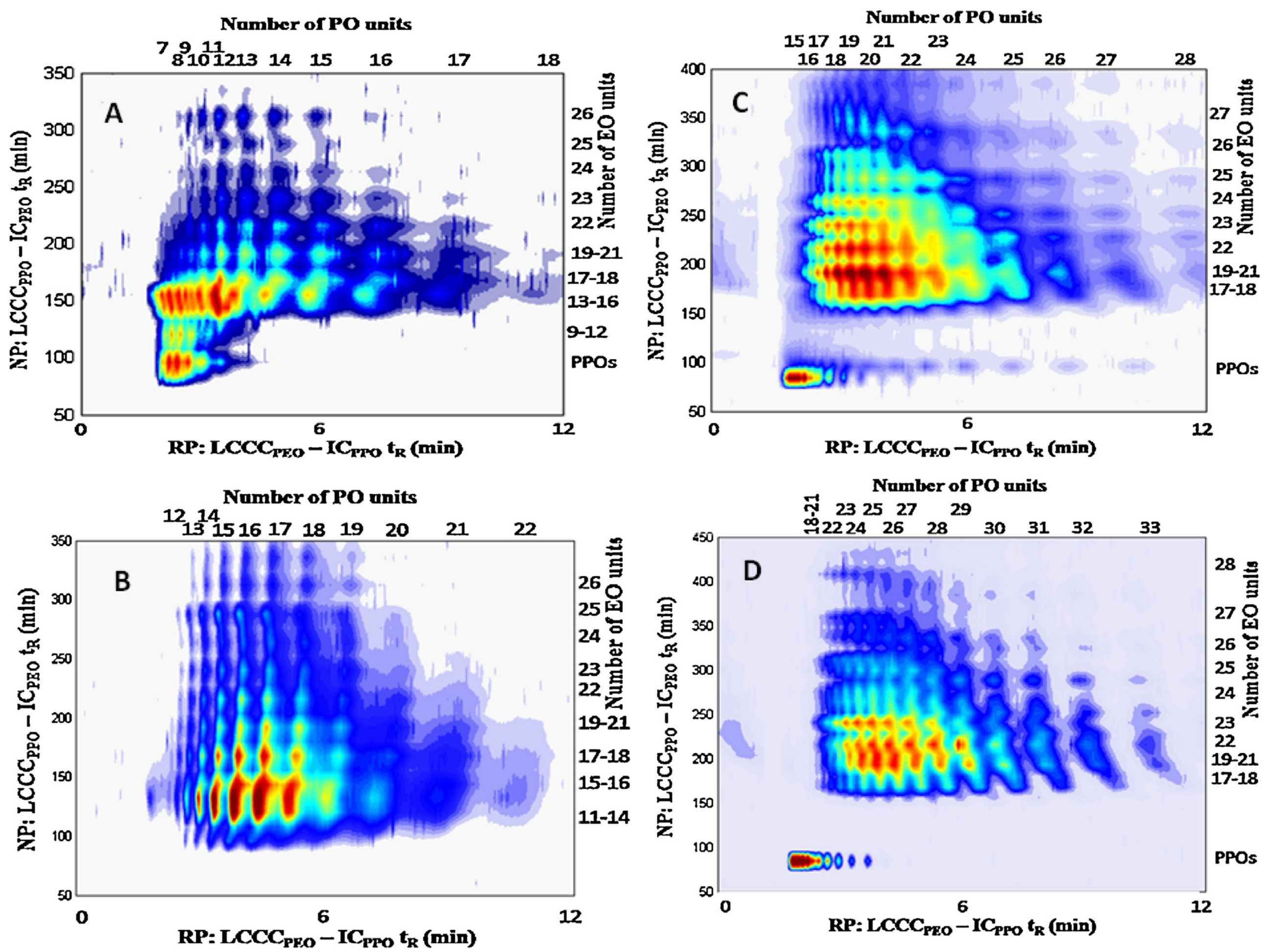

Fig. 6. 2D NPLC x RPLC contour plots, Y-axis represents first dimension: NPLC—Nucleosil, 5 μm, 300 Å 250 × 4.6 mm column in acetonitrile/water = 84/16 v/v at 0.05 mL/min; X-axis represents second dimension: RPLC—Nucleosil C18, 5 μm, 500 Å 30 + 100 × 4.6 mm column in methanol-water mobile phase (A) Pluronic-10R5 (2nd dimension conditions- methanol/water = 75/25 v/v at 1.0 mL/min, column oven temperature = 45 °C), (B) Pluronic-L35 (2nd dimension conditions- methanol/water = 80/20 v/v at 1.0 mL/min, column oven temperature = 40 °C) (C) Pluronic-1740 (2nd dimension conditions-methanol/water = 84/16 v/v at 1.1 mL/min, column oven temperature = 40 °C) (D) Pluronic-17R4 (2nd dimension conditions- methanol/water = 87/13 v/v at 1.1 mL/min, column oven temperature = 45 °C).

v/v, and 87/13 v/v are used for the second dimension analysis, in the same order. The flow rate in the second dimension was kept 1.0 mL/min for pluronic-10R5 and pluronic-L35 while 1.1 mL/min for pluronic-1740 and pluronic-17R4. The second dimension analysis is completed in 12 min. This means, the first dimension eluent should have been collected for the similar time period. It is accomplished by using two switching valves. First switching valve is equiped with two 100 μL loops that are filled alternatively by first dimension eluent in two minutes at the flow rate of 0.05 mL/min. The second pump flushes one of the loop while other is being filled. The second pump directs the flow to second switching valve that is equiped with two short trapping columns. The analyte adsorbs strongly on the stationary phase in weak mobile phase used by second pump (methanol/water = 50/50 v/v). This process is repeated six times before the second 10-port valve switches and analysis in the second dimension of the trapped polymer is accomplished while trapping of next six switches of first 10-port valve are being conducted on the second trapping column. The method allows comprehensive analysis of all the effluent from the first dimension. The effective sorption of the sample on the stationary phase in the solvent delivered by second pump and desorption by second dimension solvent (third pump) is the key aspect for IC × IC LC system. As can be seen, all products are separated with regard to both EO and PO block lengths. PPO homopolymers that is separated at void volume of column in first dimension showed oligomeric separation in the second dimension. The products with PPO as outer block have significantly more amount of PPO homopolymers compared to the product with inner PPO block. In the former case, EO should be polymerized first and used as macroinitiator for polymerization of PO whereas it is reversed for the latter. The presence of PPO homopolymers is due to the chain transfer reaction to the monomer that is more likely to happen in PO polymerization [21]. The number of PO units given at the upper x-axis and number of EO units at the right y-axis are assigned with the help of MALDI-TOF MS analysis. The 2D-LC contour plots reveal the relative abundance of EO and PO units in the block copolymers.

4. Conclusions

In this study, oligomeric separation of commercial poloxamers is reported. RP gradient LCCC allowed excellent separation with regard to PO units in the triblock copolymer along CAP of PEO, fairly independent of the block sequence. NPLC at CAP of PPO separates with regard to number of EO units in the block copolymers. The effect of block sequence is somewhat more pronounced in this case, however, separation of poloxamer with inner critical block (PPO) is acceptable compared to excellent separation for the products with outer critical block (PPO). Finally, for the first time two-dimensional mapping of commercial poloxamers is presented by a comprehensive 2D-LC IC × IC method. A reasonable mapping of commercial poloxamers with regard to both molar mass and chemical composition is depicted. MALDI-TOF MS analysis of fractions has been used to confirm the degree of polymerization and number of units of EO/PO are assigned accordingly for both one and two-dimensional analysis. Mapping the two-dimensional relative abundance of the EO and PO units with samples having the same overall chemical composition clearly shows the differences between them.

Acknowledgments:

This work is supported by NRF-Korea (2015R1A2A2A01004974).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.chroma.2016.03.008.

References

[1] M.W. Edens, R.H. Whitmarsh, Applications of block copolymer surfactants, in: I.W. Hamley (Ed.), Developments in Block Copolymer Science and Technology, WileyChichester, West Sussex (England), 2004, pp. 325–340.
[2] A. Pitto-Barry, N.P.E. Barry, Pluronic® block-copolymers in medicine: from chemical and biological versatility to rationalisation and clinical advances, Polym. Chem. 5 (2014) 3291–3297.
[3] I.W. Hamley, Block Copolymers in Solution: Fundamentals and Applications, John Wiley & Sons, Ltd., Chichester, England, 2005.
[4] S.B. Darling, Directing the self-assembly of block copolymers, Prog. Polym. Sci. 32 (2007) 1152–1204.
[5] W. Batsberg, S. Ndoni, C. Trandum, S. Hvidt, Effects of poloxamer inhomogeneities on micellization in water, Macromolecules 37 (2004) 2965–2971.
[6] N. Hadjichristidis, A. Hirao, Y. Tezuka, F. Du Prez, Complex Macromolecular Architectures: Synthesis, Characterization, and Self-assembly, Wiley, 2011.
[7] W. Radke, Polymer separations by liquid interaction chromatography: principles – prospects – limitations, J. Chromaogr. A 1335 (2014) 62–79.
[8] A. Baumgaertel, E. Altuntaş, U.S. Schubert, Recent developments in the detailed characterization of polymers by multidimensional chromatography, J. Chromatogr. A 1240 (2012) 1–20.
[9] D. Berek, Two-dimensional liquid chromatography of synthetic polymers, Anal. Bioanal. Chem. 396 (2010) 421–441.
[10] M.I. Malik, H. Pasch, Novel developments in the multidimensional characterization of segmented copolymers, Prog. Polym. Sci. 39 (2014) 87–123.
[11] H. Pasch, B. Trathnigg, Multidimensional HPLC of Polymers, Springer, Berlin-Heidelberg-New York, 2013.
[12] T. Chang, Recent advances in liquid chromatography analysis of synthetic polymers, Adv. Polym. Sci. 163 (2003) 1–60.
[13] A.M. Striegel, W.W. Yau, J.J. Kirkland, D.D. Bly, Modern Size Exclusion Liquid Chromatography, John Wiley & Sons Inc., Hoboken, NJ, 2009.
[14] M.I. Malik, B. Trathnigg, C.O. Kappe, Microwave assisted synthesis and characterization of end functionalized poly(propylene oxide) as model compounds, Eur. Polym. J. 44 (2008) 144–154.
[15] I. Park, S. Park, D. Cho, T. Chang, E. Kim, K. Lee, Y.J. Kim, Effect of block copolymer chain architecture on chromatographic retention, Macromolecules 36 (2003) 8539–8543.
[16] H. Ahmed, B. Trathnigg, Characterization of poly(ethylene glycol)-b-poly(ε-caprolactone) by two-dimensional liquid chromatography, J. Sep. Sci. 32 (2009) 1390–1400.
[17] H. Lee, T. Chang, D. Lee, M. Shim, H. Ji, W. Nonidez, J. Mays, Characterization of poly(L-lactide)-block-poly(ethylene oxide)-block-poly(L-lactide) triblock copolymer by liquid chromatography at the critical condition and by MALDI-TOF mass spectrometry, Anal. Chem. 73 (2001) 1726–1732.
[18] W. Jiang, S. Khan, Y. Wang, Retention behaviors of block copolymers in liquid chromatography at the critical condition, Macromolecules 38 (2005) 7514–7520.
[19] B. Trathnigg, Characterization of amphiphilic polymers: independent analysis of blocks in poloxamers by liquid chromatography under critical conditions, Polymer 46 (2005) 9211–9223.
[20] M.I. Malik, H. Ahmed, B. Trathnigg, Liquid chromatography under critical conditions (LCCC): practical applications in the analysis of amphiphilic polymers, Anal. Bioanal. Chem. 393 (2009) 1797–1804.
[21] M.I. Malik, B. Trathnigg, C.O. Kappe, Selectivity of PEO-block-PPO diblock copolymers in the microwave-Accelerated, anionic ring-opening polymerization of propylene oxide with PEG as initiator, Macromol. Chem. Phys. 208 (2007) 2510–2524.
[22] B. Trathnigg, A.A. Gorbunov, Characterization of EO-PO block copolymers by liquid chromatography under critical conditions, Macromol. Symp. 237 (2006) 18–27.
[23] M.V. Hulst, A.V.D. Horst, W.T. Kok, P.J. Schoenmakers, Comprehensive 2-D chromatography of random and block methacrylate copolymers, J. Sep. Sci. 33 (2010) 1414–1420.
[24] M.I. Malik, B. Trathnigg, C.O. Kappe, Amphiphilic polymers based on higher alkylene oxides. Synthesis and characterization by different chromatographic techniques, J. Chromatogr. A 1216 (2009) 1167–1173.
[25] W. Lee, D.Y. Cho, T. Chang, K.J. Hanley, T.P. Lodge, Characterization of polystyrene-b-polyisoprene diblock copolymers by liquid chromatography at the chromatographic critical condition, Macromolecules 34 (2001) 2353–2358.
[26] W. Lee, S. Park, T. Chang, Liquid chromatography at the critical condition for polyisoprene using a single solvent, Anal. Chem. 73 (2001) 3884–3889.
[27] S. Lee, H. Lee, L. Thieu, Y. Jeong, T. Chang, C. Fu, Y. Zhu, Y. Wang, HPLC characterization of hydrogenous polystyrene-block-deuterated polystyrene utilizing the isotope effect, Macromolecules 46 (2013) 9114–9121.
[28] S. Park, D. Cho, J. Ryu, K. Kwon, W. Lee, T. Chang, Fractionation of block copolymers prepared by anionic polymerization into fractions exhibiting three different morphologies, Macromolecules 35 (2002) 5974–5979.
[29] M.I. Malik, T. Mahboob, S. Ahmed, Characterization of poly(2-vinylpyridine)-block-poly(methyl methacrylate) copolymers and blends of their homopolymers by liquid chromatography at critical conditions, Anal. Bioanal. Chem. 406 (2014) 6311–6317.
[30] M. Hehn, P. Sinha, H. Pasch, W. Hiller, Onflow liquid chromatography at critical conditions coupled to 1H and 2H nuclear magnetic resonance as powerful tools for the separation of poly(methylmethacrylate) according to isotopic composition, J. Chromatogr. A 1387 (2015) 69–74.
[31] M.I. Malik, B. Trathnigg, Full separation of oligomers in block copolymers of ethylene oxide and propylene oxide, J. Sep. Sci. 32 (2009) 1771–1781.
[32] H. Lee, W. Lee, T. Chang, S. Choi, D. Lee, H. Ji, W.K. Nonidez, J.W. Mays, Characterization of poly(ethylene oxide)-block-poly(L-lactide) by HPLC and maldi-ToF mass-spectrometry, Macromolecules 32 (1999) 4143–4146.
[33] K. Im, H.-W. Park, Y. Kim, B. Chung, M. Ree, T. Chang, Comprehensive two-dimensional liquid chromatography analysis of a block copolymer, Anal. Chem. 79 (2007) 1067–1072.
[34] L. Česlová, P. Jandera, P. Česla, A study of the thermodynamics of retention of block (*co*) oligomers using high-performance liquid chromatography/mass spectrometry, J. Chromatogr. A 1247 (2012) 89–98.
[35] T. Welerowicz, P. Jandera, K. Novotná, B. Buszewski, Solvent and temperature gradients in separation of synthetic oxyethylene-oxypropylene block (*co*) polymers using high-temperature liquid chromatography, J. Sep. Sci. 29 (2006) 1155–1165.
[36] P. Jandera, M. Holcapek, L. Kolarova, Retention mechanism, isocratic and gradient-elution separation and characterization of (*co*) polymers in normal-phase and reversed-phase high-performance liquid chromatography, J. Chromatogr. A 869 (2000) 65–84.
[37] M. Mlynek, W. Radke, Critical chromatography in ternary solvents, J. Chromatogr. A 1284 (2013) 112–117.
[38] B. Trathnigg, M.I. Malik, N. Pircher, S. Hayden, Liquid chromatography at critical conditions in ternary mobile phases: gradient elution along the critical line, J. Sep. Sci. 33 (2010) 2052–2059.
[39] A. Gorbunov, B. Trathnigg, Theory of liquid chromatography of mono- and difunctional macromolecules—I. Studies in the critical interaction mode, J. Chromatogr. A 955 (2002) 9–17.
[40] C. Rappel, B. Trathnigg, A. Gorbunov, Liquid chromatography of polyethylene glycol mono- and diesters: functional macromolecules or block copolymers, J. Chromatogr. A 984 (2003) 29–43.
[41] M.I. Malik, G.W. Harding, M.E. Grabowsky, H. Pasch, Two-dimensional liquid chromatography of polystyrene–polyethylene oxide block copolymers, J. Chromatogr. A 1244 (2012) 77–87.
[42] T. Šmigovec Ljubič, K. Rebolj, D. Pahovnik, N. Hadjichristidis, M. Žigon, E. Žaga, Utility of chromatographic and spectroscopic techniques for a detailed characterization of poly(styrene-b-isoprene) miktoarm star copolymers with complex architecture, Macromolecules 45 (2012) 7574–7582.
[43] S. Ahn, H. Lee, S. Lee, T. Chang, Characterization of branched polymers by comprehensive two-dimensional liquid chromatography with triple detection, Macromolecules 45 (2012) 3550–3556.
[44] S. Moyses, Two-dimensional chromatography applied to the study of the thermo-oxidative degradation of poly(styrene-b-butadiene) star block copolymers, J. Sep. Sci. 35 (2012) 1741–1747.
[45] K. Im, Y. Kim, T. Chang, K. Lee, N. Choi, Separation of branched polystyrene by comprehensive two-dimensional liquid chromatography, J. Chromatogr. A 1103 (2006) 235–242.
[46] S. Ahn, K. Im, T. Chang, P. Chambon, C.M. Fernyhough, 2D-LC characterization of comb-shaped polymers using isotope effect, Anal. Chem. 83 (2011) 4237–4242.
[47] E. Uliyanchenko, S. van der Wal, P.J. Schoenmakers, Challenges in polymer analysis by liquid chromatography, Polym. Chem. 3 (2012) 2313–2335.
[48] B. Trathnigg, C. Rappel, R. Raml, A.A. Gorbunov, Liquid exclusion-adsorption chromatography: a new technique for isocratic separation of non-ionic surfactants; V. Two-dimensional separation of fatty acid polyglycol ethers, J. Chromatogr. A 953 (2002) 89–99.
[49] O.P. Haefliger, Universal two-dimensional HPLC technique for the chemical analysis of complex surfactant mixtures, Anal. Chem. (2003) 371–378.

[50] P. Jandera, Comprehensive two-dimensional liquid chromatography—practical impacts of theoretical considerations. A review, Cent. Eur. J. Chem. 10 (2012) 844–875.
[51] R.E. Murphy, M.R. Schure, J.P. Foley, One- and two-dimensional chromatographic analysis of alcohol ethoxylates, Anal. Chem. 70 (1998) 4353–4360.
[52] P. Jandera, J. Fischer, H. Lahovská, K. Novotná, P. Česla, L. Kolářová, Two-dimensional liquid chromatography normal-phase and reversed-phase separation of (*co*) oligomers, J. Chromatogr. A 1119 (2006) 3–10.
[53] P. Jandera, M. Halama, L. Kolarova, J. Fischer, K. Novotna, Phase system selectivity and two-dimensional separations in liquid column chromatography, J. Chromatogr. A 1087 (2005) 112–123.
[54] M.I. Malik, B. Trathnigg, R. Saf, Characterization of ethylene oxide-propylene oxide block copolymers by combination of different chromatographic techniques and matrix-assisted laser desorption ionization time-of-flight mass spectroscopy, J. Chromatogr. A 1216 (2009) 6627–6635.
[55] A.A. Gorbunov, A.V. Vakhrushev, Theory of chromatographic separation of linear and star-shaped binary block-copolymers, J. Chromatogr. A 1064 (2005) 169–181.